Eric Capo, PhD
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Dr Eric Capo is an Assistant Professor at the Department of Ecology, Environment and Geoscience, Umeå University (Umeå, Sweden) since 2023. The Capo lab studies how aquatic microbial communities change across space and time, and how they functionally respond to climate change, eutrophication, deoxygenation, and mercury pollution. They apply molecular ecology tools - metabarcoding, (ancient) metagenomics, MAGs analysis, and metatranscriptomics - to the genetic information present from water columns and sediment archives, capturing a high-resolution record of microbial life.

Main research grants

2026 Research Grant funded by Göran Gustafsson foundation (stifelse för natur och miljö i Lappland). Renewal grant of "Mapping the microbial diversity in subarctic lakes (Lappland) with a drone-sampling environmental DNA sequencing approach". Applicant: Eric Capo. 96.0 kSEK (8,680 €)

2025 Research Grant funded by Göran Gustafsson foundation (stifelse för natur och miljö i Lappland). Mapping the microbial diversity in subarctic lakes (Lappland) with a drone-sampling environmental DNA sequencing approach. Applicant: Eric Capo. 99.8 kSEK (8,970 €)

2025 Research Grant funded by the Climate Impacts Research Centre (CIRC). Unraveling microbial diversity in Swedish Arctic lakes. Applicant: Eric Cap. 80 kSEK (7,190 €)

2024 Research Grant funded by SWERVE. Funding for 8 days of ship time onboard R/V Svea. Project BIOX - Exploring the biogeochemistry and microbial ecology of oxygen-depleted fjords: historical and modern insights. Applicant: Eric Capo (co-applicants S Bertilsson, E Björn, S Bonaglia, C Bunse, J Hawkes, H Filipsson, F Nascimento, C Stairs). 1.08 MSEK (97,670 €)

2023 Kempe post-doctoral fellowship. Environmental paleomicrobiology: Inferring community composition and function from metabolic fossils. Applicants: E Libby & E Capo. 696 kSEK (63,000 €)

2023 Research grant funded by the Swedish Research Council VR - Starting Grant (grant 2023-03504). Gasping for Breath: Exploring the consequences of coastal deoxygenation on microbial processes and ecosystem services. Applicant: E Capo. 4 MSEK (342,000 €).

2023 Kempe post-doctoral fellowship. Effects of the deoxygenation of coastal systems on microbial metabolisms and related services. Applicants: E Capo & E Björn. 672 kSEK (59,000 €)

2021 Severo Ochoa post-doc fellowship (ICM, Spain). Unveiling the role of Hg-cycling microorganisms in the global ocean. Applicant: E Capo. 88,000 €

2020 Research grant funded by the Ecochange research program: Adapted or replaced? Understanding the freshwater-to-brackish transition of the Baltic Sea microbiome by ancient DNA analysis. Applicants: E Capo & AF Andersson. 130 kSEK (12,000 €)

2019 Research grant funded by Anna-Greta and Holger Crafoord's Research Grant A key to unravel lake history from the sedimentary black box Sedimentary DNA to reconstruct long-term dynamics of aquatic microbial communities. Applicant: E Capo. 120 kSEK (10,600 €)

Teaching activities

2025-present Course Ecology, Bachelor in Biology, Umeå University, Sweden
2024-present Course Genetics and Evolution, Bachelor in Biology, Umeå University, Sweden
2024-present Course Climate Change, Bachelor in Biology, Umeå University, Sweden
2024-present Course Molecular Ecology and Evolution, Master in Ecology, Umeå University, Sweden

Mentoring

2026 Marta Gabrielle (visiting PhD student 2m), Kevin Sikström (Bachelor student 3m) **2025** Yu Zhao (post-doc 3 yrs), Yangzheng Liu (visiting PhD student 1yr), Manuela Grane (Master student 1 yr), Isabel Sanz-Sáez (post-doc 2 yrs), Flúvio Modolon (post-doc 2 yrs), Anton Sandström (PhD student 4 yrs), Kajsa Brändström (Master student 6m) **2024** Marissa Despins (visiting PhD student 3m), Anton Sandström (Master student 1yr), Nicola Gambardella (PhD student 3yrs), Anthony Khairallah (Master student 6m), Panagiota Xhantopoulou (visiting PhD student 3m), Baraa Rehamnia (post-doc 2yrs) **2023** Maïlys Picard (post-doc 2yrs), Sebastien Leveque (PhD student 3yrs), Meifang Zhong (PhD student 4yrs), Fredrik Olajos (PhD student 2yrs), Júlia Dordal Soriano (PhD student 5yrs) **2021** Jordan Von Eggers (visiting PhD student 2m), Ines Barrenechea Angeles (visiting PhD student 2m), Grayson Huston (visiting PhD student 1m), Elsa Mendes (Master student 6m) **2020** Charline Fonteny (Bachelor student 3m) **2016** Léandre Vasseur (BTS student 6m).

Publications

- **Book**

Capo E, Barouillet C, Smol JP. (**2023**) Tracking environmental change using lake sediments: Sedimentary DNA (Vol 6). *Springer Cham* doi: 10.1007/978-3-031-43799-1

- **Peer-reviewed publications**

48. Zhao Y, Zheng J, Zhong M, Zheng W, Qin B, Zhu K, Zhang C, Zhao Y, Zhang E, **Capo E**, Wang R (**2026**) Sedimentary DNA and fossil archives reveal long-term phytoplankton shifts and their biogeochemical influence in a subtropical lake. *Molecular Ecology*. doi:

47. Abs E, Keuschnig C, Amato P, Bowler C, **Capo E**, Chase AB, Chavez Rodriguez L, Dabengwa AN, Dussarrat T, Guzman T, Hernandez LK, Hultman J, Küsel K, Li Z, Mankowski A, Riley WJ, Saleska S, Wingate L (**2026**) Ideas and perspectives: Using meta-omics to unravel biogeochemical changes from cell to planetary scales. *Biogeosciences*. doi: 10.5194/bg-23-5205-2026

46. Modolon F, **Capo E**, Wardle DA (**2026**) Long-term ecosystem development and retrogression drive microbial specialization for complex organic matter degradation. *ISME Communications* doi: 10.1093/ismeco/ycag157

45. Cabello AN, Salles S, Domínguez-Huerta G, **Capo E**, Camarena-Gómez TM, García-Gómez C, Sánchez A, Mangot J-F, Cerezo I, Bautista R, Pérez P, García R, Ruiz JM, Mercado JM, Ferrera I (**2026**) Environmental disturbances and cyanobacterial traits shape prokaryotic dynamics in a eutrophic Mediterranean coastal lagoon. *Environmental Microbiome*. doi: 10.1186/s40793-026-00893-9

44. Yan D, Han Y, Rioual P, Lan J, Zan S, Tang Y, Zhang H, Guo M, **Capo E** (**2026**) Shifting patterns in tropical lake phytoplankton communities coinciding with anthropogenic nutrient and pollutant trends. *Palaeogeography, Palaeoclimatology, Palaeoecology* doi: 10.1016/j.palaeo.2026.113623

43. Khairallah A, Zhong M, Dordal Soriano J, Gambardella N, Sanz-Sáez, Yan D, Bertilsson S, Björn E, Bravo AG, **Capo E** (**2026**). Low-oxygen freshwaters as ecological niches for mercury methylators. *Water Research* doi: 10.1016/j.watres.2025.125014

42. Barouillet C, **Capo E**, Debroas D, Jenny J-P, Sabatier P, Domaizon I (**2026**). Multifaceted biodiversity changes of lake microeukaryote communities in response to nutrient enrichment and climate change. *Oikos.* doi: 10.1002/oik.11616

41. Zhong M, Barrenechea Angeles I, More K, Picard M, Bertilsson S, Bravo AG, Coolen MJL**, Capo E** (**2025**). Climate-driven deoxygenation promoted potential mercury methylators in the past Black Sea water column. *Nature Water*. doi: 10.1038/s44221-025-00526-4

40. Keck F, Peller T, Alther R, Barouillet C, Blackman R, **Capo E**, Chonova T, Couton M, Fehlinger L, Kirschner D, Knüsel M, Muneret L, Oester R. Tapolczai K, Zhang H, Altermatt F (**2025**). The global human impact on biodiversity. *Nature*. doi: 10.1038/s41586-025-08752-2

39. Hao YY, **Capo E**, Yang Z, Wen S, Hu ZC, Feng J, Huang Q, Gu B, Liu YR (**2025**) Distribution and environmental preference of potential mercury methylators in paddy soils across China *Environmental Science & Technology* doi: 10.1021/acs.est.4c05242

38. Yan D, Han Y, Liu J, Zan S, Lu Y, An Z, **Capo E** (**2024**) Metagenomic analysis of sedimentary archives reveals 'historical' antibiotic resistance genes diversity increased over recent decades in the environment. *Environmental Research Letters* doi: 10.1088/1748-9326/ad850a

37. **Capo E***, Picard M*, Nakane K, Kuwae M, Bertilsson S, Kagami M, Liu X, Sakai Y, Tsugeki N (**2024**) A sedimentary DNA perspective about the influence of environmental and food web changes on the microbial eukaryotic community of Lake Biwa. *Freshwater Biology* doi: 10.1111/fwb.14326

36. Sanz-Sáez I, Bravo AG, Ferri M, Carreras J-M, Sánchez O, Sebastian M, Ruiz-González C, **Capo E**, Duarte CM, Gasol JM, Sánchez P, Acinas SG (**2024**) Microorganisms involved in methylmercury demethylation and mercury reduction are widely distributed and active in the bathypelagic deep ocean waters. *Environmental Science & Technology* doi: 10.1021/acs.est.4c00663

35. Yan D, Han Y, Zhong M, Wen H, An Z, **Capo E** (**2024**) Historical trajectories of antibiotics resistance genes assessed through sedimentary DNA analysis of a subtropical eutrophic lake. *Environmental International* doi: 10.1016/j.envint.2024.108654

34. Lin Q, Zhang K, Giguet-Covex C, Arnaud F, McGowan S, Gielly L, **Capo E**, Huang S, Ficetola GF, Shen J, Dearing JA, Meadows ME (**2024**) Transient social-ecological dynamics reveal signals of decoupling in a highly disturbed Anthropocene landscape. *PNAS* doi: 10.1073/pnas.2321303121

33. Yan D, Picard M, Han Y, An Z, Lei D, Zhao X, Zhang L, **Capo E** (**2024**) Sedimentary DNA reveals phytoplankton diversity loss in a deep maar lake during the Anthropocene. *Limnology & Oceanography* doi: 10.1002/lno.12562

32. Von Eggers J, Wisnoki NI, Calder JW, **Capo E**, Groff DV, Krist A, Shuman B (**2024**) Environmental filtering governs consistent vertical zonation in sedimentary microbial communities across disconnected mountain lakes. *Environmental Microbiology*. doi: 10.1111/1462-2920.16607

31. Yan D, Han Y, An Z, Lei D, Zhao X, Zhao H, Liu J, **Capo E** (**2024**) Anthropogenic drivers accelerate the changes of lake microbial eukaryotic communities over the past 160 years. *Quaternary Science Reviews*. doi: 10.1016/j.quascirev.2024.108535

30. Yan D, An Z, **Capo E** (**2024**) Organic matter content and source is associated with the depth-dependent distribution of prokaryotes in lake sediments. *Freshwater Biology* doi: 10.1111/fwb.14223

29. Rincón-Tomás B, Lanzén A, Sánchez P, Estupiñán M, Sanz-Saéz I, Elisabete Bilbao M, Rojo D, Mendibil I, Pérez-Cruz C, Ferri M, **Capo E**, Abad-Recio IL, Amouroux D, Bertilsson S, Sánchez O, Acinas SG, Alonso-Sáez L (**2024**) Revisiting the mercury cycle in marine sediments: a potential multifaceted role for Desulfobacterota. *Journal of Hazardous Materials* doi: 10.1016/j.jhazmat.2023.133120

28. Huston G, Lopez MLD, Cheng Y, King L, Duxbury LC, Picard M, Thomson-Laing G, Myler E, Caren C, Helbing CC, Kinnison MT, Saros JE, Gregory-Eaves I, Monchamp M-E, Wood SA, Armbrecht L, Ficetola GF, Kurte L, Von Eggers J, Brahney J, Parent G, Sakata MK, Doi H, **Capo E** (**2023**) Detection of fish sedimentary DNA in aquatic systems: A review of methodological challenges and future opportunities. *Environmental DNA* doi: 10.1002/edn3.467

27. Cabrol L*, **Capo E*,** van Vliet D, von Meijenfeldt, Bertilsson S, Villanueva L, Sánchez-Andrea I, Björn E, Bravo AG, Heimbürger-Boavida L-E. (**2023**) Redox gradient shapes the abundance and diversity of mercury-methylating microorganisms along the water column of the Black Sea. *mSystems* (* joint first authors) doi: 10.1128/msystems.00537-23

26. Williams J, Spanbauer T, Heintzman P, Blois J, **Capo E**, Goring S, Monchamp M-E, Parducci L, Von Eggers J, Alsos IG, Bowler C, Coolen M, Cullen N, Crump S, Epp L, Fernandez-Guerra A, Grimm E, Herzschuh U, Mereghetti A, Meyer M, Nota K, Pedersen MW, Perez V, Shapiro B, Stoof-Leichsenring K, Wood J. (**2023**) Strengthening global-change science by integrating aeDNA with paleoecoinformatics. *Trends in Ecology & Evolution* doi: 10.1016/j.tree.2023.04.016

25. Lin Q, Ke Z, McGowan S, Huang S, Xue Q, **Capo E**, Zhang C, Zhao C, Shen J. (**2023**) Characterization of lacustrine Harmful Algal Blooms using multiple biomarkers: historical processes, driving synergy and ecological shifts. *Water Research* doi: 10.1016/j.watres.2023.119916

24. **Capo E***, Cosio C*, Gascón Díez E, Loizeau J-L, Mendes E, Adatte T, Franzenburg S, Bravo AG. (**2023**) Anaerobic mercury methylators inhabit sinking particles of oxic water columns. *Water Research.* (* joint first authors) doi: 10.1016/j.watres.2022.119368%3C/p%3E

23. Barouillet C, Monchamp M-E, Bertilsson S, Brasell K, Domaizon I, Epp LS, Ibrahim A, Mejbel H, Nwosu EC, Pearman JK, Picard M, Thomson-Laing G, Tsugeki N, Von Eggers J, Gregory-Eaves I, Pick F, Wood S, **Capo E**. (**2023**) Investigating the effects of anthropogenic stressors on lake biota using sedimentary DNA. *Freshwater Biology* doi: 10.1111/fwb.14027

22. Zhong M*, **Capo E***, Zhang H, Hu H, Wang Z, Tian W, Huang T, Bertilsson S. (**2023**) Homogenization of water and sediment bacterial communities in a shallow lake (Lake Balihe, China). *Freshwater Biology* (* joint first author) doi: 10.1111/fwb.14016

21. **Capo E***, Feng*, Bravo AG, Bertilsson S, Soerensen A, Pinhassi J, Buck M, Karlsson C, Hawkes J, Björn E. (**2022**) Expression levels of hgcAB genes and mercury availability jointly explain methylmercury formation in stratified brackish waters. *Environmental Science & Technology* (* joint first author) doi: 10.1021/acs.est.2c03784

20. **Capo E**, Peterson BD, Kim M, Jones DS, Acinas SG, Amyot M, Bertilsson S, Björn E, Buck M, Cosio C, Elias D, Gilmour C, Goñi Urriza MS, Gu B, Lin H, Liu Y-R, McMahon K, Moreau JW, Pinhassi J, Podar M, Puente- Sánchez, Sánchez P, Storck V, Tada Y, Vigneron A, Walsh D, Vandewalle-Capo M, Bravo AG, Gionfriddo C (**2023**). A consensus protocol for the recovery of mercury methylation genes from metagenomes. *Molecular Ecology Resources* doi: 10.1111/1755-0998.13687

19. **Capo E***, Monchamp M-E*, Coolen MJL, Domaizon I, Armbrecht L, Bertilsson S. (**2022**) Environmental paleomicrobiology: using DNA preserved in aquatic sediments to its full potential. *Environmental Microbiology* (* joint first authors) doi: 10.1111/1462-2920.15913

18. **Capo E***, Broman E*, Bonaglia S, Bravo AG, Bertilsson S, Soerensen A, Pinhassi J, Lundin D, Buck M, Hall POJ Nascimento FJA, Björn E (**2022**). Oxygen-deficient water zones in the Baltic Sea promote uncharacterized Hg methylating microbial guilds in the sediment. *Limnology and Oceanography* (* joint first author). doi: 10.1002/lno.11981

17. Lin Q, Zhang K, McGowan S, **Capo E,** Shen J (**2021**) Synergistic impacts of anthropogenic nutrient and climate change on long-term water quality and ecological dynamics in contrasting shallow-lake zones. *Limnology and Oceanography* doi: 10.1002/lno.11878

16. Figueroa D*, **Capo E***, Lindh MV, Rowe OF, Paczkowska J, Pinhassi J, Andersson A (**2021**) Terrestrial dissolved organic matter inflow drives temporal dynamics of the bacterial community of a subarctic estuary (northern Baltic Sea). *Environmental Microbiology* (* joint first authors) doi: 10.1111/1462-2920.15597

15. Mattson L, Sörenson E, **Capo E**, Farnelid H, Hirwa M, Lindehoff E, Olofsson M, Svensson F, Legrand C. (**2021**) Functional diversity facilitates stability under environmental changes in an outdoor microalgal cultivation system. *Frontiers in Bioengineering and Biotechnology* doi: 10.3389/fbioe.2021.651895

14. **Capo E**, Ninnes S, Domaizon I, Bertilsson S, Bigler C, Wang X-R, Bindler R, Rydberg J. (**2021**) Landscape setting drives the microbial eukaryotic community structure in four Swedish mountain lakes over the Holocene. *Microorganisms* doi: 10.3390/microorganisms9020355

13. **Capo E**, Giguet-Covex C, Rouillard A, Nota K, Heintzman PD, Vuillemin A, Ariztegui D, Arnaud F, Belle S, Bertilsson S, Bigler C, Bindler R, Brown AG, Clarke CL, Crump S, Debroas D, Englund G, Ficetola FG, Garner R, Gauthier J, Gregory-Eaves I, Heinecke L, Herzschuh U, Ibrahim A, Kisand V, Kjær KH, Lammers Y, Littlefair J, Messager E, Monchamp M-E, Olajos F, Orsi W, Pedersen MW, Rijal D, Rydberg J, Spanbauer T, Stoof-Leichsenring K, Taberlet P, Talas L, Thomas C, Walsh D, Wang Y, Willerslev E, Woerkom A, Zimmermann H, Coolen MJL, Epp L, Domaizon I, Alsos I, Parducci L. (**2021**) Lake sedimentary DNA research on past terrestrial and aquatic biodiversity: Overview and recommendations. *Quaternary* doi: 10.3390/quat4010006

12. Sörenson E, **Capo E,** Farnelid H, Lindehoff E, Legrand C. (**2021**) Temperature stress induces shift from co-existence to competition for organic carbon in microalgae-bacterial photobioreactor community – enabling continuous production of microalgal biomass. *Frontiers in Microbiology* doi: 10.3389/fmicb.2021.607601

11. **Capo E**, Spong G, Koizumi S, Puts I, Olajos F, Königsson H, Karlsson J, Byström P (**2021**). Droplet digital PCR applied to environmental DNA, a promising method to estimate fish population abundance from humic-rich aquatic ecosystems. *Environmental DNA* doi: 10.1002/edn3.115

10. **Capo E,** Bravo AG, Soerensen A, Bertilsson S, Pinhassi J, Feng C, Andersson AF, Buck M, Björn E (**2020**) Deltaproteobacteria and Spirochaetes-like bacteria are abundant putative mercury methylators in oxygen-deficient water and marine particles in the Baltic Sea. *Frontiers in Microbiology* doi: 10.3389/fmicb.2020.574080

9. **Capo E**, Spong G, Königsson H, Byström P. (**2020**) Effects of filtration methods and water volume on the quantification of brown trout (*Salmo trutta*) and Arctic char (*Salvelinus alpinus*) eDNA concentrations via droplet digital PCR. *Environmental DNA* doi: 10.1002/edn3.52

8. Ibrahim A, **Capo E**, Wessels M, Martin I, Meyer A, Schleheck D, Epp L. (**2021**) Anthropogenic impact on the historical phytoplankton community of Lake Constance reconstructed by multimarker analysis of sediment-core environmental DNA. *Molecular Ecology* doi: 10.1111/mec.15696

7. **Capo E**, Spong G, Norman S, Königsson H, Bartels P, Byström P. (**2019**) Droplet digital PCR assays for the quantification of brown trout (*Salmo trutta*) and Arctic char (*Salvelinus alpinus*) from environmental DNA collected in the water of mountain lakes. *PLOS ONE* doi: 10.1371/journal.pone.0226638

6. **Capo E***, Rydberg J*, Tolu J*, Domaizon I, Debroas D, Bindler R, Bigler C. (**2019**) How does environmental inter-annual variability shape aquatic microbial communities? A 40-year annual record of sedimentary DNA from a boreal lake (Nylandssjön, Sweden). *Frontiers in Ecology and Evolution* (* joint first authors). doi: 10.3389/fevo.2019.00245

5. **Capo E***, Domaizon I*, Maier D, Debroas D, Bigler C. (**2017**) To what extent is the DNA of microbial eukaryotes modified during burying into lake sediments? A repeat–coring approach on annually laminated sediments. *Journal of Paleolimnology* (* joint first auth). doi: 10.1007/s10933-017-0005-9

4. **Capo E,** Debroas D, Arnaud F, Perga ME, Chardon C, Domaizon I. (**2017**) Tracking a century of changes in microbial eukaryotic diversity in lakes driven by nutrient enrichment and climate warming. *Environmental Microbiology* doi: 10.1111/1462-2920.13815

3. Domaizon I, Winegardner A, **Capo E,** Gauthier J, Gregory-Eaves I. (**2017**) Application of DNA-based methods in paleolimnology: new opportunities for investigating long-term dynamics of lacustrine biodiversity. *Journal of Paleolimnology* doi: 10.1007/s10933-017-9958-y

2. **Capo E,** Debroas D, Arnaud F, Guillemot T, Bichet V, Millet L, Gauthier E, Massa C, Develle A-L, Pignol C, Lejzerowicz F, Domaizon I. (**2016**) Long term dynamics in microbial eukaryotes communities: a paleolimnological view based on sedimentary DNA. *Molecular Ecology* doi: 10.1111/mec.13893

1. **Capo E,** Debroas D, Arnaud F, Domaizon I. (**2015**). Is Planktonic diversity well recorded in sedimentary DNA? Toward the reconstruction of past protistan diversity. *Microbial Ecology* doi: 10.1007/s00248-015-0627-2

Other publications

- **Preprints**

Abs E, Keuschnig C, Amato P, Bowler C, **Capo E**, Chase AB, Chavez Rodriguez L, Dabengwa AN, Dussarrat T, Guzman T, Hernandez LK, Hultman J, Küsel K, Li Z, Mankowski A, Riley WJ, Saleska S, Wingate L (**2026**) Ideas and perspectives: Using meta-omics to unravel biogeochemical changes from cell to planetary scales. EGUsphere

Cabello AN, Salles S, Domínguez-Huerta G, **Capo E**, Camarena-Gómez TM, García-Gómez C, Sánchez A, Mangot J-F, Cerezo I, Bautista R, Pérez P, García R, Ruiz JM, Mercado JM, Ferrera I (**2026**) Dynamics and functional traits of prokaryotic communities in a Mediterranean coastal lagoon heavily impacted by eutrophication. bioRxiv

Capo E, Peterson BD, Kim M, Jones DS, Acinas SG, Amyot M, Bertilsson S, Björn E, Buck M, Cosio C, Elias D, Gilmour C, Goñi Urriza MS, Gu B, Lin H, Liu Y-R, McMahon K, Moreau JW, Pinhassi J, Podar M, Puente- Sánchez, Sánchez P, Storck V, Tada Y, Vigneron A, Walsh D, Vandewalle-Capo M, Bravo AG, Gionfriddo C (**2022**). A consensus protocol for the recovery of mercury methylation genes from metagenomes. bioRxiv

Capo E*, Feng*, Bravo AG, Bertilsson S, Soerensen A, Pinhassi J, Buck M, Karlsson C, Hawkes J, Björn E. (**2022**) Abundance and expression of *hgc* genes and mercury availability jointly explain methylmercury formation in stratified brackish waters. bioRxiv (* joint first author)

Capo E, Bravo AG, Soerensen A, Bertilsson S, Pinhassi J, Feng C, Andersson AF, Buck M, Björn E (**2020**) Marine snow as a habitat for microbial mercury methylators in the Baltic Sea. bioRxiv

- **Chapter of books**

Barouillet C, Domaizon I, **Capo E** (**2024**). Protist DNA from lake sediments. Chapter in the book "Tracking Environmental change using lake sediments: Sedimentary DNA (Vol. 6). Springer Cham (eds Capo E, Barouillet C, Smol JP). doi: 10.1007/978-3-031-43799-1_6

Giguet-Covex C, Jelavić S, Foucher A, Morlock MA, Wood SA, Augustijns F, Domaizon I, Gielly L, **Capo E** (**2024**). The sources and fates of lake sedimentary DNA. Chapter in the book "Tracking Environmental change using lake sediments: Sedimentary DNA (Vol. 6). Springer Cham (eds Capo E, Barouillet C, Smol JP). doi: 10.1007/978-3-031-43799-1_2

Capo E, Barouillet C, Smol JP (**2024**). Using lake sedimentary DNA to reconstruct biodiversity changes. Chapter in the book "Tracking Environmental change using lake sediments: Sedimentary DNA (Vol. 6). Springer Cham (eds Capo E, Barouillet C, Smol JP). doi: 10.1007/978-3-031-43799-1_1

Vandewalle-Capo M, **Capo E**, Rehamnia B, Sheldrake M, Lee N (**2020**). The biotechnological potential of yeast from extreme habitats. Chapter in the book "Biotechnological Applications of Extremophilic Microorganisms (Life in Extreme Environments), Editor: N Lee.

Domaizon I, Arnaud F, **Capo E,** Debroas D, (**2016**) L'ADN des archives sédimentaires : une opportunité unique de retracer les changements de biodiversité lacustre. In book: De la reconstitution des paysages à l'histoire des sociétés - 10000 ans d'archives sédimentaires en zone humides, Editors: Eds, Ph Barral, M Magny, M Thiviet

- **Chapter of PhD thesis**

Capo E, Giguet-Covex C, Rouillard A, Nota K, Heintzman PD, Vuillemin A, Ariztegui D, Arnaud F, Belle S, Bertilsson S, Bigler C, Bindler R, Brown AG, Clarke CL, Crump S, Debroas D, Englund G, Ficetola FG, Garner R, Gauthier J, Gregory-Eaves I, Heinecke L, Herzschuh U, Ibrahim A, Kisand V, Kjær KH, Lammers Y, Littlefair J, Messager E, Monchamp M-E, Olajos F, Orsi W, Pedersen MW, Rijal D, Rydberg J, Spanbauer T, Stoof-Leichsenring K, Taberlet P, Talas L, Thomas C, Walsh D, Wang Y, Willerslev E, Woerkom A, Zimmermann H, Coolen MJL, Epp L, Domaizon I, Alsos I, Parducci L. (**2021**) Lake sedimentary DNA research on past terrestrial and aquatic biodiversity: Overview and recommendations. PhD thesis of Kevin Nota. Uppsala University, Department of Ecology and Genetics, Plant Ecology and Evolution. Uppsala, Sweden

Sörenson E, **Capo E**, Farnelid H, Lindehoff E, Legrand C (**2020**) Temperature stress induce shift from coexistence to competition for organic carbon in microalgae-bacterial photobioreactor community – enabling continuous production of microalgal biomass. PhD Thesis of Eva Sörenson. Linnaeus University, Department of Biology and Environmental Science. Kalmar, Sweden.

Mattson L, Sörenson E, **Capo E**, Olofsson M, Hirwa M, Svensson F, Farnelid H, Lindehoff E, Legrand C (**2020**) Functional diversity facilitates resilience to environmental changes in long-term microalgal cultivation system. PhD Thesis of Eva Sörenson. Linnaeus University, Department of Biology and Environmental Science. Kalmar, Sweden.

Winegardner A, **Capo E,** Domaizon I, Debroas D, Hajibabei M, Shokralla S, White L, Wing B, Beissner B., Gregory-Eaves (**2016**). Eukaryotic biodiversity dynamics during the Anthropocene from a mining region: an exploration using NGS. PhD Thesis of Amanda Winegardner. In thesis Aquatic biodiversity patterns along gradients of multiple stressors and disturbance histories: Integration of paleoecology and molecular techniques. MgGill University, Department of Biology, Gregory-Eaves lab, Montréal, Canada.

Figueroa D, **Capo E,** Lindh M, Rowe O, Paczkowska J, Pinhassi J, Andersson A (**2016**), Coupling between bacterial community composition and allochthonous organic matter in a sub-arctic estuary. PhD Thesis of Daniela Figueora. In thesis "Bacterioplankton in the Baltic Sea: influence of allochthonous organic matter and salinity". Umeå University, Faculty of Science and Technology, Department of Ecology and Environmental Sciences. Umeå, Sweden.

Oral presentations

Capo E. Exploring microbial life of aquatic ecosystems with environmental DNA. Seminar at NIGLAS, 24 August **2026**, Nanjing, China. Invited speaker

Capo E. MicroMates: How can we educate kids without them noticing? ISME20, 16-21 August **2026**, Auckland, New Zealand. Invited speaker

Fournier C, **Capo E**, Libby E. Zhong M. A combination of empirical data and metabolic simulation to study microbial community structure and dynamics. ISME20, 16-21 August **2026**, Auckland, New Zealand.

Leveque S, Burdett H, **Capo E**, Claverie T, Lourdes M-G Kamenos N. Bleaching driven dynamics of coral-Symbiodiniaceae-environment interactions, ICRS 2026, 21 June **2026**, Auckland, New Zealand.

Capo E, Zhong M, Fournier C, Rehamnia B, Brändström K, Andrén E, Andrén T, Andersson A, Coolen MJL, Filipsson HL, Björn E, Bravo AB, Bertilsson S. Analysing past and modern microbial data to better understand deoxygenated coastal systems. Ocean Science Meeting, 22-27 Feb **2026**, Glasgow, UK.

Tsugeki N, **Capo E**, Jiang Q, Ochi N, Kagami M, Doi H, Hayakawa K, Kuwae M. Assessment of environmental stressors on food web dynamics: evidence from sedimentary records in ancient Lake Biwa. Ecology Society of Japan meeting ESJ73, 12 March **2026**, Kyoto, Japan

Capo E. Unraveling microbial diversity in Swedish lakes. IceLab Pitch lunch. 25 May **2025**. Umeå, Sweden

Cossart T, Dordal Soriano J, Zhong M, Bratthäll T, Capo E, Bravo AG, Björn E. Methylmercury formation in an ice-covered fjord system with redox stratified water column. EGU General Assembly Conference. 2 May **2025**, Vienna, Austria & Online.

Capo E. Tracking microbial processes change related to coastal system´s deoxygenation. Ecochange meeting. 04 June **2024**. Norbbyn, Sweden

Capo E. Water deoxygenation changing microbial services in past, present and future coastal systems. IceLab Pitch lunch. 10 April **2024**. Umeå, Sweden

Capo E. Past and present deoxygenation of water columns and related impacts on microbial ecosystem services. AMRI Hand-all meeting. 08 November **2023**. Uppsala, Sweden (invited speaker)

Capo E. Studying microbial communities from past and present aquatic systems with environmental DNA. KBC Days, 07 November **2023**. Umeå, Sweden (invited speaker)

Capo E. Recovering the genetic signal of microorganisms from water columns and sedimentary archives. Seminar at the Department of Ecology and Environmental Sciences. 20 September **2023.** Umeå, Sweden

Dordal Soriano J, Bravo AG, Ruiz-González C, Pachiadaki M, Edcomb V, **Capo E**. Sinking particles in anoxic layers of Cariaco Basin as hotspots for mercury methylators. 1st Joint International Conference ICOBTE & ICHMET (2023), 6 Sep **2023**, Wuppertal, Germany

Capo E. Environmental paleomicrobiology: an emerging research field to study the long-term changes in aquatic microbial communities. Seminar at University of Ottawa (Canada). 21 June **2023** virtual meeting (invited speaker).

Bravo AG, Sanz-Saéz I, Carreras J-M, Sánchez O, Sebastián M, Ruiz-González C, **Capo E**, Gasol JM, Duarte CM, Sánchez P, Acinas S. Environmental drivers shaping the active and widespread communities involved in MeHg degradation in the deep ocean. ASLO meeting 2023, 4-9 June **2023**, Palma de Mallorca, Spain

Capo E. Microbial functional genes recovered with metagenomics from sedimentary archives. Potsdam sedDNA meeting, Potsdam, Germany. 6 June **2023** (keynote speaker)

Tsugeki N, **Capo E**, Ochi N, Kagami M, Doi H, Hayakawa K, Kuwae M. Sedimentary records revealed plankton dynamics driven by water level management, climate change and nutrient conditions in deep Lake Biwa. The 86th Annual Meeting of the Japanese Society of Limnology. Sep 19, **2022**

Von Eggers J, Wisnoki N, Calder JW, **Capo E**, Dulcinea G, Krist A, Shuman B. Deterministic processes drive microbial assembly of highly isolated disjunct mountain lake sediments. IPA-IAL **2022**.

Feng C, **Capo E**, Bravo AG, Bertilsson S, Soerensen AL, Pinhassi J, Buck M, Karlsson C, Hawkes J, Björn E. Methylmercury formation is controlled by mercury speciation and abundance of Hg methylators in stratified brackish waters of Baltic Sea, Goldschmidt2022, 15 July **2022**, Honolulu, Hawai'i, USA

Capo E. Applying molecular tools to unravel microbial transformations of mercury in aquatic and terrestrial ecosystems. Workshop Hg-Omics, 25 May **2022**, Buchillon, Switzerland (invited speaker)

Capo E., Bertilsson S, Nakane K, Kagami M, Kuwae M, Tsugeki N. Environmental-driven changes in the microbial eukaryotic community of Lake Biwa over the last 120 years. JASM2022, 19 May **2022**, Grands Rapids, Michigan, USA

Capo E. Sedimentary DNA to reconstruct past ecosystem changes in lakes and watersheds. Seminar at the Université du Québec en Abitibi-Témiscamingue (Canada), 8 March **2022**, virtual meeting (invited speaker).

Capo E. Merging two worlds: application of molecular ecology tools in paleolimnology to study the long-term changes in aquatic microbial communities. Seminar at the Department of Biology, Queen´s University (Canada), 1 March **2022**, virtual meeting (invited speaker)

Capo E. A full-year of trying to recover sediment ancient metagenomes from the Baltic Sea: torn between fun and despair. Seminar at the sedaDNA scientific society, 16 December **2021**, virtual meeting.

Capo E. Unveiling the role of Hg-cycling microorganisms in the global ocean**.** Seminar at the Department of Marine Biology and Oceanography, Institute of Marine Sciences, Spanish National Research Council (CSIC), Barcelona, Spain, 01 December **2021**, real-life meeting

Capo E. Applying molecular methods to natural environmental archives to reconstruct past life of aquatic microbial communities. Seminar at the UMR EPOC, University of Bordeaux, 8 November **2021**, virtual meeting.

Capo E. Applying sedimentary DNA approach to inform about the effects of long-term environmental changes on aquatic microbial communities. Seminar at the Department of Earth Sciences, University of Birmingham (UK), 25 October **2021**, virtual meeting (invited speaker)

Caiyan F & **Capo E.** Abundance and expression of *hgc* genes and mercury availability jointly explain methylmercury formation in stratified brackish waters. Seminar at the Department of Chemistry, University of Umeå, Sweden, 18 October **2021**, virtual meeting (joint presentation).

Capo E. Sedimentary DNA as a tool to reconstruct past aquatic microbial life. Seminar at the Department of Marine Biology and Oceanography, Institute of Marine Sciences, Spanish National Research Council (CSIC), Barcelona, Spain, 13 October **2021**, real-life meeting

Capo E. Recovering sediment ancient metagenomes: failures and successes. Mini-symposium "Recovering DNA from sedimentary archives" Uppsala Sweden 4 October **2021**, real-life meeting!

Capo E. Recovering DNA from aquatic sedimentary archives: towards understanding the long-term consequences of environmental perturbations on ecosystems. Mangrove Microbiome Initiative Meeting. 13 September **2021**, virtual meeting (invited speaker)

Capo E, Coolen MJL, Monchamp ME, Domaizon I, Bertilsson S. Decadal to millennial time series of aquatic microbial communities recovered from sedimentary DNA archives: State-of-the-art, limitations and perspectives, ASLO2021, 22-27 June **2021**, virtual meeting

Capo E. Gathering knowledge about Hg-cycling microorganisms by digging into environmental genomic data. Seminar at IVM seminar from SLU Uppsala, 05 May **2021**, virtual meeting

Capo E, Alsos IG, Ariztegui D, Armbrecht L, Barrenechea I, Belle S, Bertilsson S, Bigler C, Bindler R, Blahed I-M, Blois J, Bowler C, Boyer F, Clarke CL, Coissac E, Coolen MJL, Cordier T, Crump S, de Schepper S, Doi H, Domaizon I, Dulias K, Edwards M, Ekram MA-E, Epp LS, Everett R, Fernandez-Guerra A, Ficetola GF, Garces-Pastor S, Garner R, Gauthier J, Giguet-Covex C, Gregory-Eaves I, Harðardóttir S, Harning D, Heinecke L, Heintzman PD, Herzschuh U, Kisand V, Kuwae M, Lejzerowicz F, Littlefair J, Messager E, Meyer R, Monchamp M-E, More K, Murchie T, Nota K, Olajos F, Orlando L, Orsi W, Parducci L, Pawlowska J, Pawlowski J, Pedersen MW, Perez V, Poulain A, Ray JL, Ribeiro S, Rijal DP, Rouillard A, Rydberg J, Schulte L, Seeber P, Shapiro B, Siano R, Spanbauer, Stoof-Leichsenring K, Suchora M, Taberlet P, Talas L, Thomas C, Tsugeki N, von Eggers J, Vuillemin A, Walsh D, Wang Y, Weiner A, Williams JW, Wood J, Zimmermann H. Unifying the sedaDNA scientific community. EGU General Assembly 2021 - EGU21-4697. 28 April **2021**, virtual meeting

Williams J, Also IG, Blois J, Boyer F, **Capo E**, Clarke CL, Coolen MJL, Crump S, Edwards M, Epp LS, Fernandez-Guerra A, Goring S, Grimm E, Heintzman PD, Herzschuh U, Johnson M, Mereghetti A, Meyer R, Monchamp M-E, Nota K, Parducci L, Pedersen MW, Perez V, Rouillard A, Seeber P, Shapiro B, Spanbauer T, Stoof-Leichsenring K, Von Eggers J, Wood J, Yates J. Building cyberinfrastructure systems to support integrative, macroscale analyses of sedimentary ancient DNA records: current resources, needs and opportunities. EGU General Assembly 2021 - EGU21-4697. 28 April **2021**, virtual meeting

Capo E. The sedaDNA scientific society. GDR French Environmental Genomics "DNA et al. in paleoenvironments", 08 April **2021,** virtual meeting

Capo E. Hg in aquatic environments: Adopting omics to unveil the role of microorganisms in Hg-cycling processes. Seminar at Institute of Marine Sciences ICM-CSIC Barcelona, 08 April **2021** virtual meeting

Capo E, Feng C, Broman E, Bravo AG, Soerensen A, S Bertilsson, J Pinhassi, Buck M, Nascimento JA, Bonaglia S, Björn E. *The mercury methylation potential of pelagic redoxclines and oxygen-deficient sediment from the Baltic Sea*. Ecochange conference 24-25 November **2020**, virtual meeting

Andersson A, Figueroa D, Brugel S, **Capo E**, Zhao L, Huseby S. *Coastal phytoplankton characterization for assessing ecological state – DNA analysis usable in monitoring?* Ecochange conference 24-25 November **2020**, virtual meeting

Capo E. Using sedaDNA tool to answer unresolved questions in aquatic microbial ecology. sedaDNA cyberinfrastructure. 02 June **2020**, virtual meeting

Capo E, Buck M, Bertilsson S, Pinhassi J, Soerensen A, Andersson AF, Bravo AG, Feng C, Björn E. *Mercury methylation by microbial communities from Baltic Sea waters*. Ecochange conference 21-22 November **2019**, Kalmar, Sweden

Capo E, Bertilsson S, Pinhassi J, Buck M, Feng C, Soerensen A, Björn E. *Mercury microbiology in oxygen deficient coastal seas*, NENUN10, 25-27 September **2019**, Uppsala, Sweden

Capo E, Ninnes S, Rydberg J, Bigler C, Domaizon I, Debroas D, Wang X-R, Bindler R. *A glance at the history of aquatic microbial communities: a sedimentary-DNA based metabarcoding approach to characterize 10,000 years records from Swedish mountain lakes*, Symposium of Aquatic Microbial Ecology (SAME) 16, 1-6 September **2019**, Potsdam, Germany

Andersson A, Figueroa D, Brugel S, **Capo E**, Zhao L, Huseby S. *Molecular identification of marine phytoplankton classes – usable in monitoring?* Baltic Sea Science Congress (BSSC) 2019, 19-23 August **2019**, Stockholm, Sweden

Capo E, *Application of environmental DNA analysis to track biodiversity in northern lakes***.** Seminar at Department of Ecology and Environmental Science, 03 April **2019**, Umeå University, Sweden

Capo E, Rydberg J, Tolu J, Domaizon I, Debroas D, Bindler R, Bigler C. *How does climate-driven weather variability impact a lake ecosystem? Tracking changes in microbial communities in a 40-year varved record from an ice-covered boreal lake using sedimentary* DNA. IPA-IAL joint meeting 2018, 18-21 June **2018**, Stockholm, Sweden

Capo E, Spong G, Königsson H, Norman S, Byström P, *Estimating fish abundance in lakes via environmental DNA analysis***.** Seminar at Department of Ecology and Environmental Science, Umeå University, 08 November **2017**, Sweden

Domaizon I, **Capo E**, Debroas D, Arnaud F, Perga ME, Millet L, Guillemot T, Gauthier, Winegardner A, Gregory-Eaves I. *High-throughput sequencing of sedimentary DNA: new opportunities for investigating long-term dynamics of lacustrine biodiversity driven by local and global forcing factors*. SIL Austria meeting 2017, 26-28 October **2017**, Innsbruck, Austria

Capo E, Debroas D, Arnaud F, Bigler C, Domaizon I. *Sedimentary DNA based-approaches to study regime shifts in lacustrine trophic networks related to anthropogenic and climate pressures: the case of microbial eukaryotic communities*. CIRC Symposium, 26-28 September **2017**, Abisko, Sweden

Capo E, Debroas D, Arnaud F & Domaizon I. *Tracking a century of changes in microbial eukaryotic diversity driven by eutrophication and climate warming*. 07 Mars **2017**, Seminar at Laboratoire d'Ecologie Microbienne (UMR 5557 Lyon, France)

Capo E, Debroas D, Arnaud F, Perga M-E, Chardon C, Domaizon I. *Tracking a century of change in lacustrine microbial eukaryotes driven by local and global forcings*. ISME16 Symposium. 21-26 August **2016**. Montreal, Canada

Figueroa D, Lindh M, **Capo E,** Lundin D, Pinhassi J, Andersson A. *Bacterioplankton community response to allochthonous organic matter discharges in coastal areas of the Bothnian Bay*. Swedish Society for Marine Sciences 18-20 November **2015**, Lund, Sweden

Winegardner A, **Capo E,** Domaizon I, Debroas D, Hajibabei M, Shokralla S, White L, Wing B, Beisner B, Gregory-Eaves I. *The use of Next-Generation Sequencing to characterize eukaryote biodiversity in* *lakes with historical metal pollution*. Quebec Centre for Biodiversity Science Annual Symposium. 29-30 October **2015**. Montreal, Canada

Capo E, Arnaud F, Bichet V, Chardon C, Debroas D, Develle A-L, Guillemot T, Lejzerowicz F, Domaizon I. *Response of lacustrine protistan assemblages to climatic and anthropogenic pressures: a paleolimnological view based on sedimentary DNA*. Symposium for European Fresh Water Sciences SEFS9, 6 July **2015**, Geneva, Switzerland

Capo E, Debroas D, Arnaud F & Domaizon I. *Paleo-ecological approach of lacustrine microbial diversity*. Seminar at EMG Department, Umeå University, 18 February **2015**, Umeå, Sweden

Domaizon I, Debroas D, Bronner G, Taïb N, Lepère C, Mangot JF, Savichtcheva O, **Capo E**, Lyautey E, Chardon C, Jacas L, Arnaud F, Pignol C, Perga ME, Gregory Eaves I. *Application of DNA-based methods for investigating modern and past diversity of planktonic taxa in lakes: Planktonic biodiversity and its forcing factors*. In: DNAWatch2013, 4-5 December **2013**, Frasnes, France

Posters

Capo E, Zhong M. Unraveling microbial diversity in Swedish lakes with drones. ISME20, 16-21 August 2026, Auckland, New Zealand.

Zhong M, Sanz Saéz I, Dordal Soriano J, Despins M, Lamborg C, Bertilsson S, Björn E, Garcia Bravo A, **Capo E**. Eco-evolutionary regimes of mercury methylation in the global ocean. ISME20, 16-21 August 2026, Auckland, New Zealand.

Sanz-Saéz I, Gambardella N, Azaroff A, Bravo AG, Dordal-Soriano J, Lavergne C, Lopez-Garcia E-M, Moreau JW, Sorel Ngou J, Peireira-Garcia C, Wang Y, Zhao L, **Capo E**. merAB-DB: A consensus database for a standardised detection of methylmercury demethylation and mercury reduction genes. Goldschmidt 2026, 13 July **2026**, Montréal, Canada.

Dordal Soriano J, Zhong M, Cossart T, Sanz-Sáez I, Brathäll T, Xanthopoulou P, Ruiz-González C, Björn E, **Capo E**, Bravo AG, Distribution of putative mercury methylators along environmental gradients in a redox-stratified brackish system, Goldschmidt 2026, 13 July **2026**, Montréal, Canada.

Zhong M, Dordal Soriano J, Picard M, Cossart T, Bertilsson S, Alsos IG, Schomacker A, Kjellman S, Bravo AG, Björn E, **Capo E**. Present and past microbial metabolisms in the oxygen-deficient coastal system Rossfjordvatnet, Norway", UCMR Days 22 January **2025**, Aula Nordica, Umeå University, Umeå, Sweden

Dordal Soriano J, Ruiz-González C, Royo-Llonch M, Zhong M, Pachiadaki M, Edcomb V, Bravo AG, **Capo E**. Upwelling events enhance mercury methylation gene expression in marine sulfidic waters. XVIII Symposium on Aquatic Microbial Ecology (SAME18), Aquatic Microbiomes on a Changing Planet, 28 September - 3 October **2025**, Barcelona

Ferrera I, Salles S, Cabello A, Domínguez-Huerta, **Capo E**, Garciá-Gómez C, Camarena-Gómez MT, García R, Mercado J. Shifts in microbial communities in a Mediterranean hypersaline coastal lagoon after an ecosystem disruptive algal bloom. ISME19 International Symposium on Microbial Ecology, Cape Town, South Africa, Aug **2024**.

Capo E, Peterson B, Jones D, Storck V, Liu YR, Kim M, Lin H, Amyot M, Acinas S, Bertilsson S, Björn E, Bowman K, Buck M, Cosio C, Elias D, Gu B, Lamborg C, Pinhassi J, Pachiadaki M, Podar M, Tada Y, Vandewalle-Capo M, Walsh D, Moreau J, McMahon K, Gilmour C, Bravo AG, Gionfriddo C. Towards building a consensus protocol for the recovery of the genes involved in mercury methylation (h*gcAB*) from environmental genomic data. Ocean Science Meeting 2022, 27 Feb-4 March **2022**. virtual.

Spanbauer T and **Capo E**. A ~4,000-year record of the microbial eukaryotic community from Yellowstone Lake (Yellowstone National Park, USA). AGU Fall meeting 2019. 9-13 December 2019, San Francisco, USA

Ibrahim A, **Capo E**, Straile D, Meyer A, Epp L, Schlebeck D. Lake Constance: Reconstructing the Eukaryotic community using sedhDNA. INQUA 2019. 25-31 July 2019, Dublin, Ireland

Figueroa D, **Capo E**, Lindh M, Rowe O, Paczkowska J, Pinhassi J, Andersson A. Changes of bacterial community's structure: responses to environmental disturbance and ADOM discharge in a subarctic estuary. SAME 15, Symposium on Aquatic Microbial Ecology. 3-8 September 2017, Zagreb, Croatia

Figueroa D, **Capo E**, Lindh M, Rowe O, Paczkowska J, Pinhassi J, Andersson. Influence of allochthonous organic matter on bacterioplankton community composition in a subarctic estuary in the northern Baltic Sea. SAME 14, Symposium on Aquatic Microbial Ecology. 23-28 August 2015, Uppsala, Sweden

Capo E., Debroas D., Arnaud F., Keck F. & Domaizon I., Is planktonic diversity well recorded in sedimentary DNA? Toward the reconstruction of past protistan diversity. Journée des doctorants ED SISEO, 4 December 2014, Annecy, France

Capo E., Debroas D., Arnaud F., Pignol C. Taib N. Bronner G. Domaizon I. Paleoecological study of lake microbial diversity: long term dynamic of protists and cyanobacteria diversity related to environmental factors. DNAWatch2013, 4-5 December 2013, Frasnes, France

Domaizon I., Savichtcheva O., **Capo E**., Arnaud F., Debroas D., Villar C., Pignol C., Jenny J.P., Alric B., Perga M.E. Dynamique et diversité à long terme des cyanobactéries révélées par l'ADN préservé dans les archives sédimentaires du lac du Bourget. ArcheoPal2013, 14-16 October 2013, Frasnes, France

Demuez M., Zhao C., Fall S., Roussel F., Effantin G., **Capo E**., Héry M., Rodrigue A., Navarro E. and Mandrand-Berthelot M.A. Role in nickel homeostasis of genes isolated from the bacterial metagenome of an ultramafic soil. Metals Homeostasis 5th International IMBG Meeting. 17-21 September 2012, Autrans, France

Education and past appointments

2021-2023 Post-doc at the Institute of Marine Sciences, ICM-CSIC, Barcelona, Spain
2020-2021 Guest at Dpt. Aquatic Sciences and Assessment, SLU, Uppsala, Sweden
2019-2021 Post-doc at Dpt. Chemistry, Umeå University, Umeå, Sweden
2017-2019 Post-doc at Dpt. Ecology and Environmental Sciences, Umeå University, Umeå, Sweden
2013-2016 PhD student at UMR CARTTEL, INRA, Thonon-les-Bains, France
2011-2013 Master student in the Master of Microbial Ecology, Lyon 1 University, France
2010-2011 Bachelor student in the Bachelor of Microbiology, Lyon 1 University, France
2007-2009 HND (DUT) student in Biological and Biochemical Analyzes, IUT Tours, France

Editorial activity

2021-2022: Guest Editor of the special issue "Interpreting environmental and evolutionary change in freshwater systems using DNA preserved in sediments" in the journal *Freshwater Biology*

Since 2020: Review Editor for the journal *Freshwater Biology*

Since 2017: Pre-publications reviews of > 30 manuscripts for: Freshwater Biology (4), Frontiers in Microbiology (2), Scientific Reports (2), Environmental Science & Technology Letters (2), Journal of Paleolimnology (2), Environmental DNA (2), Quaternary Science Reviews (1), Nature Communications (1), Proceedings of the Royal Society B (1), ISMEJ (1), Environmental Microbiology (1), Current Biology (1), Molecular Ecology (1), Communications Biology (1), Frontiers in Marine Sciences (1), Molecular Ecology Resources (1), BMC Microbiology (1), Geomicrobiology Journal (1), Environmental Science and Pollution Research (1), L&O Letters (1).

Press Releases

2025 Climate clues and microbial mysteries – Svea's first SWERVE expedition completed – link
2025 Exploring the Hidden Microbial Dynamics of the Swedish West Coast - link
2025 Researchers' card game teaches children about microorganisms - link
2025 New documentary on the challenges of climate research - link
2025 Climate change may increase the spread of neurotoxin in the oceans - link
2025 The devastating impact of humans on biodiversity - link

Events, outreach and others

Co-creator of the card game MicroMates™, shown at three outreach events (Forskarfredag 2024-2025, Pint of Science 2025) – see article.

Co-convener of the **SIL2022** session 'Studying the long-term dynamic of freshwater ecosystems through sedimentary DNA research: a potential tool for management and conservation', 07-10 August 2022, Berlin, Germany

Co-convener of the **JASM2022 session** 'Deciphering past aquatic ecosystem dynamics using sedimentary ancient DNA', 14-20 May 2022, Grand Rapids, Michigan, USA.

Lead convener of the **OSM2022 session CT11** 'Mercury transformations in marine ecosystems', 27 Feb-4 March 2022, Honolulu, HA, USA

Co-organizer of the **Mini-symposium "Recovering DNA from sedimentary archives"**, 4th October 2021, Uppsala, Sweden

Co-leader of Past Global Changes **Working Group PaleoEcoGen**, July 2021

Co-organizer of the **1st symposium** of the African sedaDNA working group, 9-10th June 2021, virtual

Co-creator of the **African sedaDNA working group** (July 2021 > 25 members)

Co-convener of the **EGU21 session BG2.6** Reconstruction of past ecosystems using sedimentary ancient DNA: Applications, method developments, analytic pipelines and data management, 19-30 April 2021, virtual

Founder and Coordinator of the **sedaDNA scientific society** (February 2021 > 260 members)

Creator and PI of the **Meta-Hg working group** (July 2020 > 50 members)

Creator of the newsletter **Sedimentary-DNA** (April 2019, > 350 subscribers)

Part of the Local Organizing Committee of **SEFS9 conference** (Geneva 2015)

Part of the organizing board of PhD student day (doctoral School SISEO, 2015)

Contribution to the science fairs at INRA UMR CARRTEL (Thonon-les-Bains, 2015)

Scientific skills

Field cruises	Swedish lakes (2017 & 2018), Baltic Sea (2019 & 2020), Norwegian fjords (2024), Swedish fjords (2025)
Molecular Biology	DNA/RNA extraction, ancient DNA extraction, PCR, clone libraries, primers design, qPCR, ddPCR (QX200), metabarcoding, metagenomics, metatranscriptomics
Data analysis	α, β, γ diversity, network topology, statistics, bioinformatics
Tools	R software (incl. coding), PAST, Seaview, MEGA, BioEdit, BLASTn, Gephi, Cytoscape, panam2, mothur, qiime2, vsearch, cutadapt, bowtie2, bbmap, prodigal, hmmer, megahit, samtools, metabat2, checkM, gtdbtk

Microscopy	Phytoplankton counting
Microbiology	Microbial culture, species identification, API Gallery
Chemistry	CG-MS, SPIR, hydrocarbons and lipids extraction
Sampling	Water and sediments sampling, filtration (peristaltic & vacuum pump)
Languages	French (Native Language); English (Proficient), Swedish, (Basic), Chinese (Basic)
Other	Project development, experimental design, budget management, mentoring

Other research Grants

2024 Research Grant funded by Agencia Estatal de Investigación (AEI) for 2024 Proyectos de Generación de Conocimiento call - PID2024-163029NB-I00. Project Play-Mer: Players, drivers and niches of methylMercury formation in marine water columns in a context of increased oxygen depletion. Applicant: Andrea Garcia Bravo (Eric Capo as collaborator). 181,000 €

2024 Research Grant funded by Kempe foundation. Funding for an anaerobe glovebox dedicated to biogeochemistry and microbiology research work on trace metals. Applicant: Erik Björn (Eric Capo as collaborator). **0.6 MSEK**.

2023 Research Grant funded by PN2022 -Consolidación Investigadora-Subprograma Estatal de Incorporación- Programa Estatal para Desarrollar, Atraer y Retener Talento - PEICTI 2021-2023. Microbial and Organic Matter drivers of mercury biogeochemical cycling in Marine particles (MOMa). Applicant: Andrea Garcia Bravo (Eric Capo as collaborator). 157,820 €

2023 Research Grant funded by Kempe foundation. Ansökan om medel till instrumentering för bestämning av spårmetallföreningar (TDGC-ICPMS) Applicant: Erik Björn (Eric Capo as collaborator). **2 MSEK**.

2023 Research grant funded by Sea Grant Pennsilvania. Applicants: J-P Balmonte & C Bove. (E Capo as collaborator) Microbial Menaces on Microplastics: A Spatiotemporal Study in the Schuylkill River Watershed. 148,100 USD (**1.7 MSEK**).

2023 Research grant funded by 2023 Endeavour Fund from New Zealand government SMARTIDEA. Applicant; J Pearman (E Capo as collaborator). Applying a functional evidence approach to prioritise lake restoration initiatives. 1 MNZD (**6.5 MSEK**)

2022: Research grant funded by the Swiss National Science Foundation (Synergia, CRSII5_213522). Deep biosphere-geosphere interactions at the top of the world (DIGESTED): An interdisciplinary approach to interpret a Myr climate record from lake Nam Co (Tibetan Plateau) Applicant: H Vogel (E Capo as collaborator, WP4). 3 MCHF (**33.5 MSEK**)

2022 Research grant funded by the Swedish Research Council VR (740731-1955), Evolution and biogeography of the brackish microbiome. Applicant: A. Andersson (E Capo as collaborator, WP3). **4 MSEK** (380,000 €)

2022 Research grant funded by Leading Academy in Marine and Environment Pollution Research at Ehime University (Lamer fund). Development of methodology using environmental DNA as a tool for assessing long-term plankton dynamics. Applicant: N Tsugeki. (E Capo as collaborator). 240,000 yens (1,750 €).

2022 Research grant funded by the Center for Ecological Research (2022jurc-cer01), Kyoto University. Development of methodology using sedimentary DNA as a tool for assessing long-term zooplankton dynamics. Applicant: N Tsugeki. (E Capo as collaborator). 300,000 yens (2,310 €).

2021 Research grant funded by Swedish Research Council for Sustainable Development Formas (grant 2021-00819). Methylmercury formation in coastal seas: understanding risks associated with permanent and temporary oxygen deficiency. Lead applicant: E. Björn (E Capo as co-applicant). 3 MSEK (290,000 €)

2021 Research grant (2021-2022) funded by Leading Academy in Marine and Environment Pollution Research at Ehime University (Lamer fund). Development of methodology using environmental DNA as a tool for assessing long-term plankton dynamics. Applicant: N Tsugeki. (E Capo as collaborator). 250,000 yen (1,940 €).

2021 Research grant (2021-2024) funded by the Japan Society for the Promotion of Science. Grants-in-Aid for Scientific Research. Development of new method for reconstructing microorganisms using sedimentary DNA to assess the effect of global warming measures on lake ecosystems. Applicant: N Tsugeki. (E Capo as collaborator). 3,200,000 yen (10,700 €).

2020 Research grant funded by the Japan Society for the Promotion of Science: Reconstruction of host and parasite during the past century in Lake Biwa by using paleolimnological analysis. Applicant: N Tsugeki (E Capo as collaborator). 900,000 yen (6980 €).

2019 Travel Grant by SAME for SAME16 symposium (Potsdam, Germany, 1-6 sept 2019). 500 €

2018 Travel Grant funded by Wallenberg Foundation Travel Grant for the workshop “Metabarcoding of microbial communities” (Berlin, Germany). 1 060 €

2016 Travel Grant funded by AFL for ISME16 Symposium (Montréal, Canada). 400 €

2016 Travel Grant by ISME for ISME16 Symposium (Montréal, Canada). 1000 €

Internships and workshops

April 2018	**Workshop in Botanischen Museum of Berlin**, Berlin, Germany *Metabarcoding of microbial communities*
August 2014	**UNESCO International Summer School** in Limnology**,** Evian-les-bains, France *Lake ecology and temporal dynamics*
July 2013	**Summer School** at LOV UPMC, Villefranche-sur-mer, France *Marine plankton diversity*
June-July 2013	**Volunteer work** at ISARA, Lyon, France (with Joël Robin) *Phytoplankton dynamics in ponds*
Jan-May 2013	**Master 2 Internship**, EMG UMR 5005 Ecully, France (with Timothy Vogel & Catherine Larose) *Microbial degradation of hydrocarbons in snow*
Apr-Aug 2012	**Volunteer work** in “Société des Amis de Navarrosse”, Biscarrosse, France *Cyanobacterial diversity of coastal lakes*
July 2012	**Summer school** at Marine Station of Arcachon, France *Plankton and macrofauna ecology*
Jan-Apr 2012	**Master 1 Internship**, MAP UMR 5240, Villeurbanne, France (with Marie-Andrée Mandrand) *Effect of an ATPase gene, isolated from metagenomics DNA of mining spoils in New-Caledonia, on the resistance to metals*